LESSON 3.8

Solve Problems with Unit Rates

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by using unit rates to compare options.

WHY THIS MATTERS You're the new supply manager for the arcade! You need to order tokens, prizes, and snacks from different suppliers. Each supplier offers a different package deal. Your job is to find the unit rate for each deal and figure out which supplier gives you the best value for every item you need to order. Fun Supply Co. sells 500 tokens for $60. Arcade Depot sells 350 tokens for $38.50. Which supplier gives the better value for tokens?

REMEMBER

Solve Problems with Unit Rates

Solving Problems with Unit Rates. Formula: Unit Price = Cost ÷ Quantity | Total Cost = Unit Price × Quantity A unit rate is the cost for just 1 item, like the price per token. When packages have different sizes, the unit rate lets you compare them fairly.

1. Find the unit rate by dividing total cost by quantity.
2. Compare unit rates to determine the best deal.
3. Multiply unit rate by desired quantity to calculate total cost

Words: Unit Rate *(Tasa unitaria)* | Better Buy *(Mejor compra)* | Comparison *(Comparación)* | Per Unit *(Por unidad)* | Rate *(Tasa)*

1 **Circle the letter of the best answer. Show how you know.**

About how many pounds is 10 kilograms? (1 kg ≈ 2.2 lb)

(A) about 2.2 pounds
(B) about 4.5 pounds
(C) about 12.2 pounds
(D) about 22 pounds

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

About how many centimetres are in 4 inches? (1 in ≈ 2.54 cm)

(A) about 1.6 cm
(B) about 6.5 cm
(C) about 10.2 cm
(D) about 25.4 cm

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A race is 5 kilometres long. About how many miles is that? (1 mi ≈ 1.6 km)

(A) about 1.6 miles
(B) about 3.1 miles
(C) about 5 miles
(D) about 8 miles

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

You change 10 pounds into kilograms using 1 kg ≈ 2.2 lb. Should the number of kilograms be greater or less than 10?

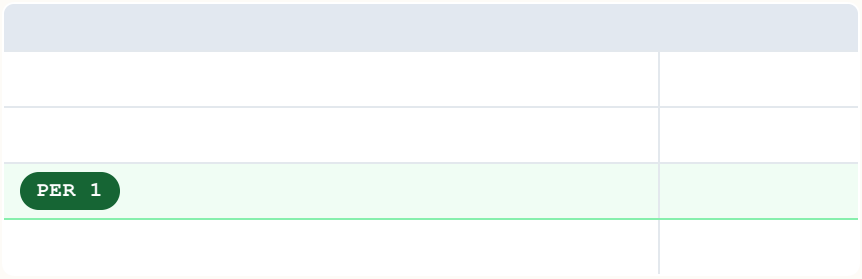
(A) Greater than 10, because a kilogram is lighter
(B) Exactly 10
(C) You cannot tell
(D) Less than 10, because a kilogram is heavier than a pound

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The arcade needs 600 prize tickets printed. Printer A prints 150 tickets in 5 minutes. Printer B prints 200 tickets in 8 minutes. The manager wants to use the faster printer. What is Printer A's unit rate?

(A) 5 tickets per minute
(B) 25 tickets per minute
(C) 30 tickets per minute
(D) 150 tickets per minute

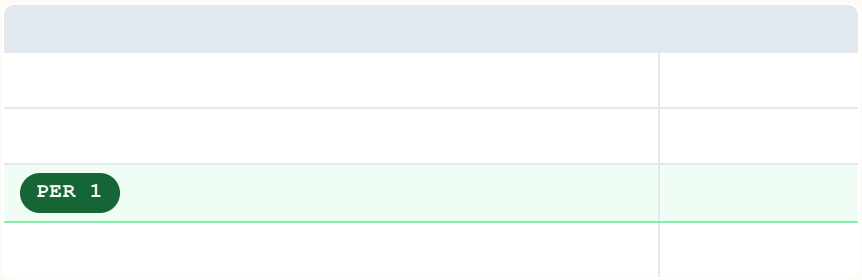


1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 **Circle the letter of the best answer. Show how you know.**

The arcade needs 600 prize tickets printed. Printer A prints 150 tickets in 5 minutes. Printer B prints 200 tickets in 8 minutes. The manager wants to use the faster printer. What is Printer B's unit rate?

(A) 8 tickets per minute
(B) 25 tickets per minute
(C) 30 tickets per minute
(D) 200 tickets per minute



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

SENTENCE STARTER I find the better buy by comparing the ___ for each option. The option with the ___ unit rate is the better buy because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

For each prize item, find the unit rate of both deals and sort each card into Better Buy or Worse Buy.

Groups: Better Buy (lower unit rate) | Worse Buy (higher unit rate)

_____ Snack packs: 4 for $5.00 ($1.25 each)

_____ Snack packs: 6 for $9.00 ($1.50 each)

_____ Stickers: 25 for $8.75 ($0.35 each)

_____ Stickers: 10 for $4.00 ($0.40 each)

_____ Tokens: 250 for $20.00 ($0.08 each)

_____ Tokens: 100 for $9.00 ($0.09 each)

2 **Write the letter of the matching item on each line.**

Match each prize deal to its unit price (price ÷ quantity).

_____ Glow bracelets: 5 for $4.50

_____ Bouncy balls: 8 for $6.00

_____ Mini erasers: 3 for $2.40

_____ Temporary tattoos: 10 for $3.50

(A) $0.90 each

(B) $0.75 each

(C) $0.80 each

(D) $0.35 each

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

The arcade needs 600 prize tickets printed. Printer A prints 150 tickets in 5 minutes. Printer B prints 200 tickets in 8 minutes. The manager wants to use the faster printer. Which printer is faster, and how long for 600 tickets?

(A) Printer A, 20 minutes

(B) Printer B, 20 minutes

(C) Printer B, 24 minutes

(D) Printer A, 24 minutes

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The snack bar compares two popcorn deals: 12 oz for $3.00 or 20 oz for $4.00. Which is the better buy?

(A) 12 oz for $3.00 — $0.25 per oz

(B) They cost the same per ounce

(C) Not enough information

(D) 20 oz for $4.00 — $0.20 per oz

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The delivery van brings arcade supplies 150 miles in 3 hours. What is the unit rate in miles per hour?

(A) 30 miles per hour

(B) 45 miles per hour

(C) 50 miles per hour

(D) 153 miles per hour

PER 1	

6 **Circle the letter of the best answer. Show how you know.**

The ticket counter sells redemption tickets two ways: 8 tickets for $2.00 or 15 tickets for $3.00. Which costs less per ticket?

(A) 8 tickets for $2.00 — $0.25 each

(B) They cost the same per ticket

(C) Cannot determine

(D) 15 tickets for $3.00 — $0.20 each

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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